

PackWise AI: Intelligent Attachment Optimization System for Doll Packaging

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The Industry Challenge

Bottlenecks in Manual Toy Attachment & Packaging Design

✗ Over-engineered Fasteners

Excessive wire ties, elastic loops, and XPE foam inflate raw material costs & create unnecessary packaging waste.

✗ High Factory Labor Time

Complex manual tie configurations slow down high-speed assembly line workers, increasing cost per unit.

✗ Transit Damage & Accessory Loss

Improper attachment locations lead to product movement, accessory detachment, or damaged display pose.

✗ Subjective Trial-and-Error

Packaging decisions lack historical data analytics, resulting in lengthy revision loops between teams.

System Features & Capabilities

Comprehensive Modules Tailored for Packaging Engineers



1. Product & Pose Analysis (YOLO)

Automated detection of doll pose complexity, joint anatomical support points, and attached accessories.

2. Attachment Recommendation Engine

Suggests precise strap types (elastic loops, molded pulp, ties), optimal quantities, and exact placement coordinates.

3. Risk & Drop-Test Predictor

Simulates Master Carton 10-drop and SIOC 17-drop test outcomes, movement risk, and accessory loss risk.



4. Cost & Sustainability Optimizer

Calculates unit strap cost (\$), assembly labor time estimation, and recyclability score in real time.

System Workflow & User Roles

Role-Based Permissions & End-to-End Approval Lifecycle



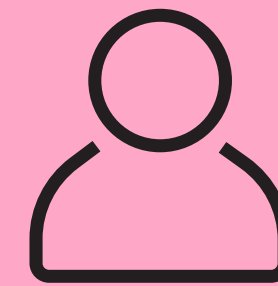
Packaging Engineer

1. Upload prototype photo & metadata.
2. Review AI attachment coordinates.
3. Run drop-test & risk simulation.
4. Submit packaging plan for review.



Product Manager

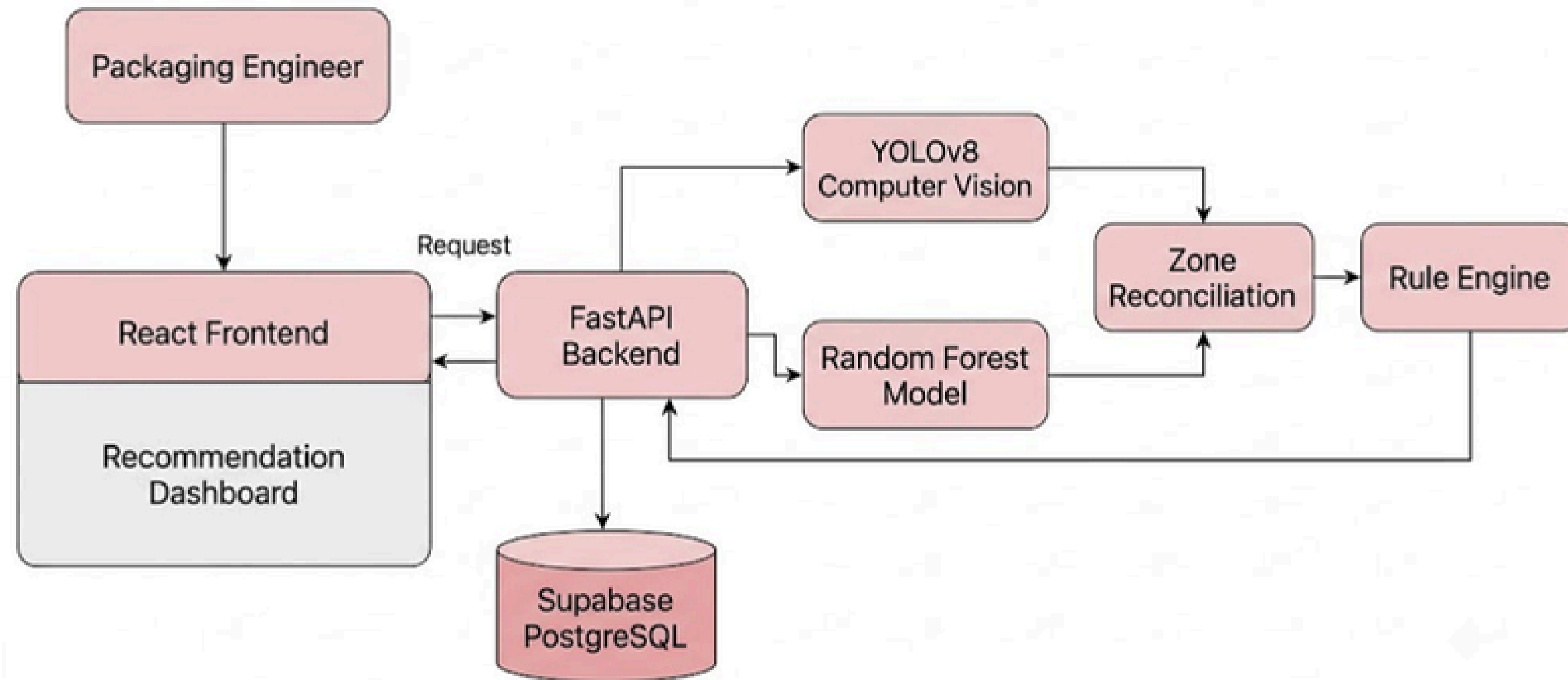
1. Receive pending approval alert.
2. Evaluate cost and sustainability impact.
3. Approve or request revision.
4. Lock final specification for production.



Administrator

1. Manage user roles & access.
2. Update master strap pricing database.
3. Configure system threshold rules.
4. Maintain audit logs.

System Architecture



Technology Stack

Infrastructure & core technologies supporting PackWise AI system architecture.

🔍 AI & ML ENGINE ×

- YOLOv8 (Ultralytics)
- Roboflow
- Random Forest & XGBoost
- Rule-Based Inference Engine
- Scikit-Learn
- OpenCV

🔍 FRONTEND ×

- React 19 & TypeScript
- TanStack Start & Router
- Vite & TailwindCSS
- Recharts
- HTML2PDF.js

🔍 DATABASE ×

- Supabase
- Supabase Auth
- Supabase Storage
- Cloudinary

🔍 BACKEND ×

- FastAPI (Python)
- REST API & CORS Middleware
- Pydantic
- Joblib & Pandas

Model Performance Benchmark: Random Forest vs XGBoost

Prediction Target	Random Forest (Accuracy %)	XGBoost (Accuracy %)	Improvement (XGB – RF)
 Head Strap	94.0%	95.0%	+1.0%
 Waist Strap	74.5%	78.0%	+3.5%
 Hand Strap	99.0%	100.0%	+1.0%
 Leg Strap	88.0%	100.0%	+12.0%
 Back Support	90.0%	94.5%	+4.5%
 Base Support	99.0%	100.0%	+1.0%
 Material	90.0%	93.0%	+3.0%
Average 	90.64%	94.07%	+3.43%



THANK YOU